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Green Campus Energy Optimization

& Smart Power Management System

1 .Project Overview

This project aims to make the university eco-friendly and sustainable by using energy optimization methods inside the campus .We must reduce energy usage to prevent worsening of the environmental quality (Yildız, 2018). This project focuses on reducing electrical energy wastage and saving energy with environmentally friendly practices along with improving campus facilities.

Purpose, Scope and Objectives

- Purpose

This project focusses on saving energy by minimizing electricity usage inside the campus by using smart lighting and modern technologies .Reducing energy usage helps reduce demand for fossil fuels which leads to lower harmful emissions and environmental damage (Lefèvre, et al., 2023)

I. Focus Areas of Project

- ❖ Minimize electric waste inside the campus
- ❖ Make campus eco-friendly
- ❖ Reduce overall university operational costs
- ❖ Use saved resources for infrastructure development
- ❖ Use solar energy

- Scope

The scope includes restructuring campus electrical systems for energy optimization, implement motion sensing lights, use of inverter ACs, installation of solar panels for sustainable solar energy, building energy consumption monitoring systems and use energy efficient equipments along with developing necessary software systems.

- **Included Areas**

❖ Classrooms

❖ Corridors

❖ Cafeteria

❖ Washrooms

❖ Recreational rooms

❖ Labs

❖ Office rooms

- **Non – Included Areas**

❖ Parking areas

❖ Buildings outside main campus facility

❖ Student residences

❖ Playground

❖ Open spaces

- Sports facilities

- Objectives

  The focus of the project is sustainable energy usage and reducing dependence on dirty (non-eco-friendly) energy. According to the International Energy Agency (IEA) (2022), the 'first fuel' for a sustainable energy system is energy efficiency

  - Reduce energy usage by 25 % in first year
  - Automate electrical components to turn off when not in use
  - Increased use of eco-friendly lighting (e.g.: add more windows for natural lighting)
  - Decrease in electrical bills

Business Case

- **Statement of Problem**

  A typical campus usually wastes about 30% to 50% of the total electricity used in a year. A large portion of this waste is due to unnecessary lighting and AC usage. Such wastage leads to a massive increase in electrical bills and subsequent increase in the campus operational costs.

- **Problem Remedy**
  - Use of energy efficient lights
  - Install motion sensors inside the campus which turn OFF the lights when people are not around
  - Use inverter ACs as they are energy efficient
  - Increased use of natural lighting wherever possible
  - Automated shutdown systems after class time

- **Benefits**
  - ❖ Energy conservation
  - ❖ Reduction in electricity bills
  - ❖ Better allocation of funds
  - ❖ Improved campus recognition and eco-friendly image

- **Long - Term Sustainability**

The energy optimization project is deeply rooted in the idea of long-term sustainability as, even though initial investments are high, the return on investment (ROI) within a few years is about double as electricity prices are rising at an alarming rate .Increasing power demands from expanding technology sector and the need to upgrade aging grid infrastructure are the main reasons of rising electricity bills (Fortune, 2026) . It also helps reduce carbon emissions .In the future solar panels can also be used as a way to further reduce dependence on dirty energy.

- **Expected ROI**

Payback Period = (1190944) ÷ 566498 = 2.1 years.

Statement of Work (SOW)

Objective : Implement energy optimization and electricity management systems to reduce electric waste inside the campus

Scope : Restructuring campus systems and use energy efficient devices to minimize electricity wastage and reduce carbon emissions

Timeline : The project is expected take about 6 months to complete

Responsibilities : Change old and inefficient electrical components along with implementing new energy efficient systems and timely reporting of work schedule .

Payment terms : 40% of payment will be done upfront for procurement of equipments and remaining 60% will be paid after completion of the project.

Change control : Project plans have to be approved by the board of directors and any changes to the plan has to be reviewed by the board before implementation.

Communication plan : Progress reports and client meetings every 2 weeks

Acceptance criteria : All systems implemented working properly and meets or exceeds expected functionality

Project Deliverables & Non – Deliverables

- **Deliverables**

No.	Deliverables	Description
1	Energy monitoring system and software	This energy monitoring system helps calculate the energy saved and consumed along with a software which controls all operations
2.	Smart energy conservation systems	These include smart motion sensing lights, inverter ACs and other power efficient electrical components
3.	System repair and operational guide	This is a manual that guides user on how to operate the new system and repair any minor problems
4.	Electricity cost reduction	By implementing energy conservation system there will be a minimum of 20% reduction in operational costs
5.	Free annual maintenance	For 3 years after handover of the project the annual maintenance fee will be free

Figure 1(By Author)

- **Non – Deliverables**
  - ❖ Free yearly maintenance of all installed systems after 3-year period
  - ❖ Free replacement of damaged items after warranty period
  - ❖ Implementation of energy management systems outside campus building

Project Success Criteria & KPI's

Project success criteria	Key performance indicator

Reduce electricity wastage inside campus	Decrease in electricity bill
All systems work without any problems	Positive feedback from students and staff
Work must be completed within 6 months	Weekly progress report and work being done before expected completion date
Implementation of all systems should not exceed budget	Cost effective planning and procurement of good quality equipments at discounted rates

Figure 2(By Author)

Stakeholders

There are two types of stakeholders involved in this project

- Primary
- Secondary

PRIMARY STAKEHOLDERS

1. Students
2. University staff
   -professors
   - registration team
   - Junior lecturers

SECONDARY STAKEHOLDERS

1. Government
2. Environmental NGO'S
3. Private sponsor companies
4. Electronic Gadget's Suppliers

3. University board members

5. Skilled workers

4. Project management team

- Manager

-Reacher team

- Finance team

- Software and IT teams

5. Contactors and technical teams.

--> These are the primary and secondary stakeholders.

Primary stakeholders are any group or individuals who can affect or is affected by the achievement of the organization's objectives (Freeman, 1984). These are also called as internal stakeholders.

Secondary stakeholders are the people or working groups, who aren't directly involved or within this project but, still they play a major role or have a major influence on this project. These are also called as external stakeholders.

- **Interest and Influence**

All the stakeholders are not the same, as different stakeholders have different levels of power/ influence and interest in and by this project. Hence, we can categories them by their power and interest on this project.

High Influence + High Interest University Board Directors Project manager Research team Environment	**High Influence + Moderate/low Interest** Government Authorities Private Sponsors NGO'S Maintaince Team Finance team Suppliers
Low influence + High interest Students University staff Local communities	**Low influence + Low interest** Part-Time workers Skilled Laborers

Figure 3(By Author)

High influence+ high interest :-

These are the people, groups and environmental factors which have the highest power and interest to influence and get affected by this project.

such as,

-> **University board Directors** – Approve or deny this project, financial support and Cost reductions in electricity.

-> **Project Manager** -- Most informed, monitors closely.

-> **Environment** – Can affect the project's success, project outcomes can reduce carbon emissions

High influence + Moderate/low influence

These are the people or working groups who hold a lot of power to influence this project but gets low/moderate impact.

Government Officials – Plays a major role in authorization and for electric supply.

Private Sponsors – Financials helps to run the project.

NGO's -- permits to run this project in the specific area.

Maintainance team – Maintains and look after each and everything once the project has successfully been installed.

Suppliers – Has a major role for essential goods, gadgets required for this project.

High Interest + low influence

These are the people or groups who gets most affected by this project.

Students and university staff – Has to adapt to new energy-saving systems.

Local communities -- Environmental impact and improved sustainability.

Low Influence + low interest

These are the working groups who holds the least/no power towards this project and doesn't get affected by this project.

Part-time workers and skilled laborers – Can be easily replaced.

Work Distribution structure

The entire work for this project has been distributed into several parts so that we can get an idea of individual work for this project.

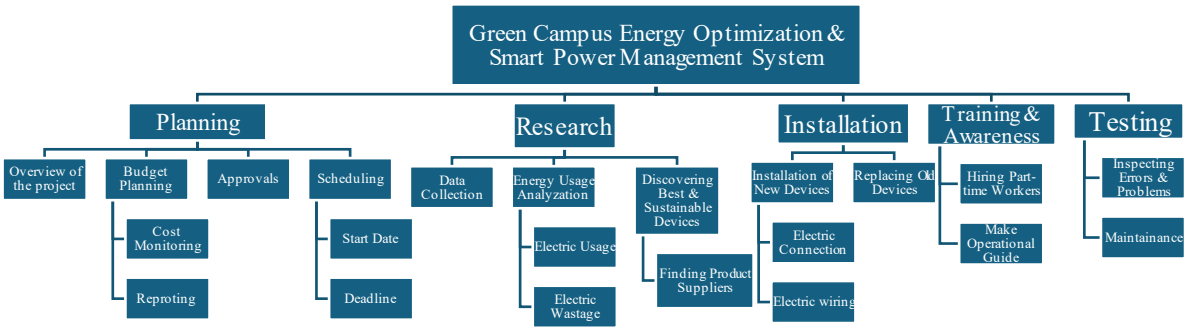


Figure 4(By Author)

Organization Breakdown Structure

Organization breakdown structure(OBS) used to show project organization and different organizational units which are responsible for doing the work (Kerzner, 2017). It helps identify

the individual work they are assigned and responsible for .This gives a clear overview of the work which must be split into different groups in this project.

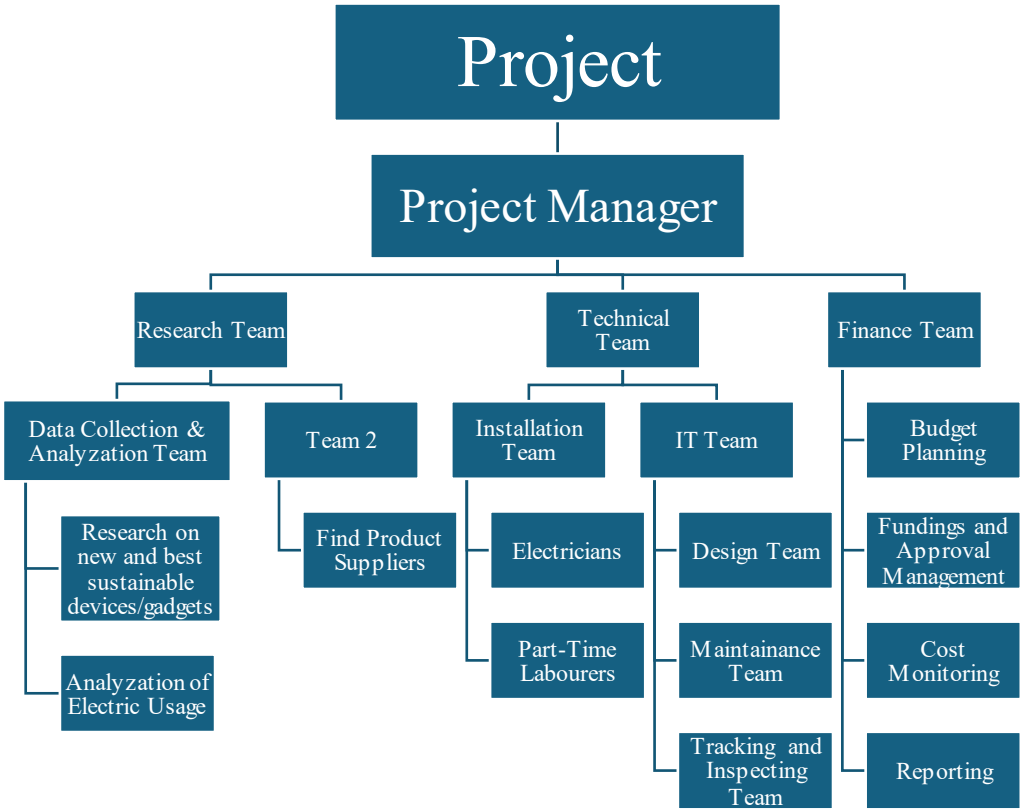


Figure 5(By Author)

All the teams and departments who are going to get involved in this project work has been listed in OBS's flow chart

Responsible Assignment Matrix

In this matrix table it's shown which part of department, or a person is responsible for each task.

It gives a clear view of how the entire work has been distributed and done, and it also shows how each and every department is responsible for every work which has been involved in this project.

The work or responsibility for each person or department will be categorized by -

- R – responsible -> The person/team doing the work.
- A – Accountable -> The person in charge of the final result
- C – Consulted --> The person/team who give advice for the work
- I – Informed --> The person who only get updated of the work

	Manger	Research team 1	Research team 2	Electricians	Part-time labours	Finance Team	IT team	Manintaince team
Planning the overview of the project	R	C	C	I	I	C	C	C
Getting approvals	R	C	C	I	N/A	C	C	C
Defining projects schedule	R	I	I	I	N/A	I	I	I
Electric usage Analyzation	A	R	C	C	N/A	N/A	N/A	I
Discovering sustainable devices	A	R	I	C	N/A	I	I	I
Finding Suppliers	A	C	R	N/A	N/A	I	N/A	N/A
Installation of new sustainable devices	I	N/A	N/A	R	R	N/A	I	I
Monitoring the work	R	C	C	N/A	N/A	C	C	C
Tracking and inspecting for any problems	I	C	N/A	C	N/A	N/A	R	I
Overall maintaince	A	N/A	N/A	R	R	N/A	I	A
Budget Planning	A	I	C	N/A	N/A	R	C	N/A
Managing fundings from sponsors	A	N/A	N/A	N/A	N/A	R	I	N/A
Cost monitorization	A	N/A	C	N/A	N/A	R	N/A	N/A
Reporting	A	R	R	C	C	R	R	R

Figure 6(By Author)

2 . Risk Assessment

Risk Identification

- Technical Risks

  The technical risks include:

  1. Sensors not working
  2. Software system Malfunction
  3. Lights and ACs malfunction
  4. Approval time took way longer than expected.
  5. Bauantrag approval
  6. VDE 0100 inspection

- Financial Risks

  1. Budget overruns
  2. Increased wages for part-workers
  3. Increase in raw material and electric component prices
  4. ROI calculation error

- Operational Risks

  1. Staff Injuries
  2. Work delays
  3. Environmental disturbance
  4. Key specialists takeoff

Risk Mitigation Strategies

- Buy sensors and electrical components with warranty
- Regular software and hardware maintenance

- To avoid budget overrun use contract based work
- To avoid staff injuries and work efficiently, give staff training
- Have a contingency budget for overruns

3 . Project Budget

- **Budget**

Equipment One-time cost

An average campus usually requires around 60 to 120 street lights per km^2. If we assume that the campus is approximately 1 km^2 with 5 lecture buildings and each building contains 3 lecture halls,15 medium sized classrooms and 20 tutorial rooms.

If lecture halls use 30 LED'S, medium classrooms use 12 LED's and tutorial rooms use 6 LED's the campus would require a total of 1,950 energy-efficient LED lights in total

Aside from these each lecture building is assumed to require 120 hall lights, thus requiring 600 hall lights in total.

359.91€ for every street light with solar.

12.48€ for every Energy-saving LED light in classroom

We can use motion sensors to turn off lights when nobody is around. Every 2 streetlights need one senso and in the buildings, every floor needs 5 sensors. Total 60 units.

We need also at least 6 main meters for those electrical appliances.

Installation cost

For installation, it's assumed every working hour costs 35€. Installing each street light takes 2 hours and 0.1 hour for each energy-saving LED light.

ELECTRIC POWER SAVING

Every kw*h electric power costs 0.18€.

When all 100w street lights are retrofitted to solar street light and it's assumed each light works 10 hours a day, we can save 1kw*h everyday for each light.

40W for normal LED lights and energy-saving light needs 30W. Motion sensors can also help save energy, but they are only installed for hall lighting and not for classroom lighting. It is assumed that lights are left on in unoccupied areas around 70% of the time.

Salvage value

The decommissioned equipment can be recycled for approximately 6480€.

Solar electric power production

It is assumed that each building has a rooftop area of approximately 1,200 m² available for potential solar panel installation. After accounting for maintenance space, shading distance, and rooftop equipment, approximately 70% of the rooftop area remains usable, resulting in an estimated usable area of 840 m². Approximately 5–6 m² of solar panels are required to generate 1 kW of electricity. So, 6 rooftops of building can produce about 840KWh of electricity per hour. The commercial solar PV cost approximately 1150€/KW. The times with sunshine are average 12hours a day and

80% days a year is sunny. So, the actual power product amount is 840×12×0.8=8064 KWh per day. The total price of commercial solar PV is 840×1150= **966,000**

ROI (ignore the staff salaries)

Payback Period = (1190944) ÷ 566498 = 2.1 years.

Item	Qty	Unit €	Total €
LED-Solar Street Light	60	359.91	21594.6
Energy-saving LED light for classroom	2550	12.48	31824
Motion sensors	60	9	540
main meters	6	1500	9000
Installation labor	375	35	13125
salvage value	N/A	N/A	-6480
commercial solar PV	840	1150	966000
Subtotal	3051	N/A	1035604
Contingency 15%	N/A	N/A	N/A
Total	N/A	N/A	1190944

Figure 7(By Author)

category	saving electric power(kw*h/day)	Price of electric power(€)	The saving money pro day(€)	per year(€)
electricity savings of street light	60	0.18	10.8	**3944.7**
motion sensors and energy-efficient light	297.6	0.18	53.568	**19565.712**
only energy-efficient light	195	0.18	35.1	**12820.275**
Commercial solar PV	8064	0.18	1451.52	**530167.68**
total	8616.6	N/A	1550.988	**566498.37**

Figure 8(By Author)

Monthly cost

Staff salary: It is assumed that the salaries of all staff are based on the average wage level.

Manager: 6000€/month

Researcher: 3500€/month

Electricians: 3200€/month

Accountant: 3200€/month

AP/AR Assistant: 2600€/month

IT technician: 3300€/month

Maintenance Technician: 3500€/month

4 . Project Schedule

- ## Scheduling

To schedule the project, we should list the steps of the project.

Phase 1: Approval

We should discuss the feasibility of our project, confirm team members, budget and design which will take 1 month.

Phase 2: Coordination

In this phase, the finance team should start their work and contact equipment suppliers and installation teams.

Manager should contact the headmaster so he can notify all staffs, students and professors in the campus.

Phase 3: Installation

When suppliers provide the first batch of equipment, installation teams can begin their work. This Optimistic timeline of installation estimates 2 months. Conservative estimate - 4 months.

Phase 4: Profit Generation

Once all the equipment is installed, the project officially enters profit generation phase.

Phase	Start	Duration	End
Approval	26-May-26	31	26-Jun-26
Coordination & Purchase	27-Jun-26	45	11-Aug-26
Installation	12-Jul-26	90	10-Oct-26
Profit Generation	11-Sep-26		

Figure 9(By Author)

GANTT CHART

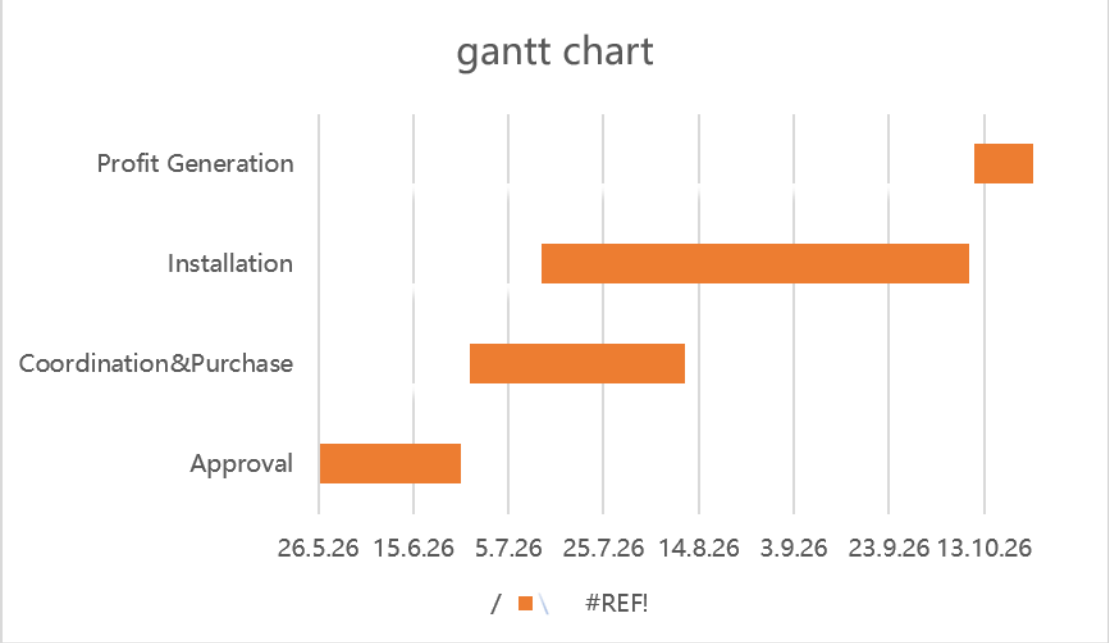


Figure 10(By Author)

5 .Tracking and Evaluation

- **Purpose of tracking:**

Tracking keeps a clear overview of the entire work going according to the planned schedule, budget and scope .According to Drucker (2008) we cannot manage something which we have not measured .It helps identify and monitor – delays, shortages, missing elements and resolve it by the end of this project. And it also helps monitor performance of the team members.

- ## Methods Used for tracking:

These are the following methods used and helped track this project:

1. Work breakdown structure and organization breakdown structure :- It gives a view on the entire work of this project and who is responsible for each work as it divides the work between individual groups, Hence it make easier to track the work of this project.

2. Responsible assignment matrix (RAM) :- It's a table which shows members roles involved in each work. It helps locate how the individual person or group has participated in each and every work.

3. Communication with team members: It gives regular updates of new works and ideas which are implemented in this project.

4. Onsite and off-site meetings :- Helps give new ideas which can be used in this project to make it more efficient by the end if this.

5. Progress monitoring :- Helps in keeping track of the scheduling of this project and timeline of each work.

6. Weekly reporting :- for keeping track on when exactly each work has started and ended.

7. Gant chart monitoring :- Timeline used to check delays and overlaps.

- ## Tracking table

	Planned Duration	Actual duration	Status	Reason if delayed
Requirement analysis (Approvals, scheduling, planning, fundings)	**4 weeks**	**5 weeks**	**Delayed**	**Got delayed in getting approvals**
Energy data collection and analysis	**3 weeks**	**4 weeks**	**Delayed**	**Huge amount of data has been found to analyze**
Research on sustainable devices	**4 weeks**	**6 weeks**	**Delayed**	**Found difficulty in finding best energy optimization devices and finding devices which are within the project budget**
Installation and replacing old devices	**3 weeks**	**3 weeks**	**Completed**	**On schedule**
Hiring workers	**2 weeks**	**2 weeks**	**Completed**	**On schedule**
Inspecting errors	**1 week**	**2 weeks**	**Delayed**	**More and complex errors have been found which took more time to solve**
Making an operational guild	**1 week**	**1 weeks**	**Completed**	**On schedule**

Figure 11(By Author)

Issues identified: 1. **Bauantrag bureaucratic delay (Activity A):** German construction supervision authority required additional documentation on the 840 kW load before approval.
2. **Infrastructure incompatibility (Activity G):** Two of the five lecture buildings had outdated wiring that was not fully compatible with the new motion sensor system, requiring unplanned rework.
3. **Supplier price renegotiation (Activity C):** Initial equipment quotes came in above the budgeted unit costs for solar street lighting, requiring the Finance Team to renegotiate terms before contracts could be signed.
4. **Failed first VDE inspection (Activity H):** The inspector flagged incomplete documentation of electrical work on the first visit, requiring a second scheduled inspection and a short hold on commissioning.

Action taken:

1. **Bauantrag delay:** Project Manager engaged the Baurechtsexperte (construction law consultant, Risk #11) to fast-track the missing documentation; weekly follow-ups were scheduled with the construction supervision office until approval was granted.
2. **Infrastructure incompatibility:** Electricians performed a targeted compatibility audit on the remaining buildings (per mitigation for Risk #7) before continuing rollout, and the budget contingency reserve (15%) was partially drawn to cover the unplanned rewiring cost.
3. **Supplier renegotiation:** Finance Team activated the backup supplier list identified during the Coordination phase mitigation planning (= (Risk #8) and secured a revised quote within budget, with prices locked under contract to prevent further increases (Risk #9).
4. **Failed VDE inspection:** Electricians conducted the pre-inspection dry run that had been planned (Risk #12), identified the missing documentation themselves on a follow-up self-check, and resubmitted complete records ahead of the second official inspection, which passed without further issues

- # Evaluation

Purpose of evaluation :-

Evaluation has been conducted to determine whether the project met its objectives, and to identify strengths and weaknesses and improvements for future projects.

Evaluation against objectives :-

- ❖ Reduce energy usage by 25 % in first year
- Partially achieved – as the measures has not yet fully determined for an entire year but the measures till now looks promising.

- ❖ Automate electrical components to turn off when not in use

☺ Achieved -- these types of lightings have been heavily brought to use.

- ❖ Increased use of eco-friendly lighting (e.g.: add more windows for natural lighting)
- Partially Achieved- usage of eco-friendly devices has brought to use within the campus.

- ❖ Decrease in electrical bills

☺ Achieved – electric bills have reduced as expected due to usage of smart lightings it saves a lot of electricity.

Strengths :-

1. A clear overview of the project before it starts.
2. Brilliant project idea which can be heavily brought to several universities.
3. Environmentally friendly project
4. Easy approvals and support from NGO'S
5. This project has a lot of potential to be continued in the future by which a high percentage of employment will be generated.

Weaknesses :-

1. delays in research work
2. Risks while electric work

Recommendations for future projects.

- Use backup scores from previous projects for research work and research work.
- improve risk management planning and bring more safety measures while working.
- Allocate extra buffer time for research and data collection of heavy tasks.
- Increase stakeholder communication to better understand their needs.

Final Outcomes :-

This project met its objectives.

Created a project which is eco-friendly which gives a lot of benefits in the future.

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